GRAVITATION — FORMULA & METHODS

Class 11 Physics | CBSE • JEE Main • NEET Exam Revision Sheet

NCERT CH. 7

1. UNIVERSAL LAW OF GRAVITATION & FIELD

MUST KNOW

Newton's Universal Law of Gravitation

EXAM FAVOURITE

$$F = G \cdot (m_1 \cdot m_2) / r^2 \quad \Rightarrow \quad \mathbf{F}_{12} = -G \cdot (m_1 m_2 / r^2)\, \hat{r}_{12}$$

Universal Constant (G): 6.674×10^{-11} N m^2 kg^{-2}

Dimension of G: $[M^{-1} L^3 T^{-2}]$

Conditions: Valid for point masses & spherically symmetric bodies.

Vector Property: Central force, obeys Newton's 3rd Law ($\mathbf{F}_{12} = -\mathbf{F}_{21}$).

Gravitational Field Intensity (E)

$$\mathbf{E} = \mathbf{F} / m = (G M / r^2)\, (-\hat{r})$$

SI Unit: N kg^{-1} or m s^{-2} | **Dim:** $[L T^{-2}]$

Note: Identical to acceleration due to gravity g.

Acceleration due to Gravity (g)

$$g = G M_E / R_E^2 \approx 9.81\ m\ s^{-2}$$

Mass of Earth (M_E): 5.97×10^{24} kg

Radius of Earth (R_E): 6.37×10^6 m

2. VARIATIONS IN ACCELERATION DUE TO GRAVITY (G)

MUST KNOW

Derivation — Variation of g with Altitude (Height h) & Depth (d)

- **At Height h:** $g(h) = G M_E / (R_E + h)^2 = g \cdot (1 + h/R_E)^{-2}$. For $h \ll R_E$, using binomial expansion: **$g(h) \approx g \cdot (1 - 2h / R_E)$**
- **At Depth d:** Mass interior to radius $r = R_E - d$ is $M' = M_E \cdot (R_E - d)^3 / R_E^3$.
  Hence, $g(d) = G M' / (R_E - d)^2 \Rightarrow$ **$g(d) = g \cdot (1 - d / R_E)$**. At Earth's center ($d = R_E$), $g = 0$.

SHORTCUT — Fractional & Percentage Changes:

• Height $h \ll R_E$: $\Delta g / g = -2h / R_E$ | % change = $(2h / R_E) \times 100\%$

• Depth d: $\Delta g / g = -d / R_E$ | % change = $(d / R_E) \times 100\%$

• Equal decrease condition: $g(h) = g(d) \Rightarrow d = 2h$ (for small heights).

CAUTION — Formula Selection Mistakes:

• Do NOT use $g(h) = g(1 - 2h/R_E)$ if $h > 0.1\ R_E$ (~640 km).
Always use the exact formula $g(h) = g \cdot [R_E / (R_E + h)]^2$.

• g is MAXIMUM at the surface of Earth and DECREASES both going upwards and downwards.

Numerical Method — Finding Weight at Height h = R_E / 2

Given: $W_{surface} = mg = 63$ N, $h = R_E/2$ → Use Exact Formula: $g' = g / (1 + h/R_E)^2$ → $g' = g / (1 + 0.5)^2 = (4/9)\, g$ → **$W' = (4/9) \times 63$ N = 28 N**

3. KEPLER'S LAWS & ANGULAR MOMENTUM CONSERVATION

MUST KNOW

Law	Mathematical Expression & Statement	Key Physical Consequence
1. Law of Orbits	Planets move in ellipses with the Sun at one focus. Perihelion = $r_P = a(1-e)$, Aphelion = $r_A = a(1+e)$.	Semi-major axis $a = (r_P + r_A) / 2$.
2. Law of Areas	Areal velocity is constant: $dA / dt = \mathbf{L} / (2m) = (r\, v \sin\theta) / 2 = constant$.	Consequence of **Conservation of Angular Momentum** ($\mathbf{L} = m\, r_P\, v_P = m\, r_A\, v_A$). Speed is highest at perihelion ($v_P > v_A$).
3. Law of Periods	$\mathbf{T^2 = (4\,\pi^2 / G\,M_S) \cdot a^3}$ $\Rightarrow T^2 \propto a^3$	Constant $k = T^2/a^3$ depends ONLY on central mass M_S, independent of planet mass.

4. GRAVITATIONAL POTENTIAL ENERGY (V) & POTENTIAL (U)

MUST KNOW

Gravitational Potential Energy (V) & Potential (U)

EXAM FAVOURITE

$$V(r) = -(G\,M\,m)/r \quad | \quad U(r) = V(r)/m = -(G\,M)/r \quad | \quad \mathbf{E} = -(dU/dr)\,\hat{r}$$

Reference: Zero potential energy defined at infinity ($r \to \infty$).

Sign Convention: Always negative for bound systems.

Work Done (W_{12}): $W = V(r_2) - V(r_1) = -G\,M\,m\,(1/r_2 - 1/r_1)$

System of N Masses: $V_{total} = -\sum_{i<j} (G\,m_i\,m_j / r_{ij})$

5. ESCAPE SPEED (V E)

EXAM FAVOURITE

Derivation — Escape Speed from Earth's Surface

- **Initial Mechanical Energy at Surface:** $E_i = K_i + V_i = (1/2)\,m\,v_e^2 - (G\,M_E\,m)/R_E$
- **At Infinity (Minimum Escape Condition):** $E_f = K_f + V_f = 0 + 0 = 0$
- **By Conservation of Energy ($E_i = E_f$):** $(1/2)\,m\,v_e^2 - (G\,M_E\,m)/R_E = 0$
- **Final Formula:** $\mathbf{v_e = \sqrt{(2\,G\,M_E / R_E)} = \sqrt{(2\,g\,R_E)}}$ **≈ 11.2 km s^{-1}** (for Earth)

SPECIAL CASES & SHORTCUTS:

• **Moon's Escape Speed:** $v_e \approx 2.3\ km\ s^{-1}$. Thermal velocity of gas molecules exceeds this, explaining why the Moon has *no atmosphere*.

• **Body projected with $v = n\,v_e$ (n > 1):** Speed far away in interstellar space is $\mathbf{v' = v_e\,\sqrt{(n^2 - 1)}}$.

CAUTION — Common Assumptions & Errors:

• Escape speed is **INDEPENDENT** of mass of projected body (m) and angle of projection (θ).

• Escape speed **DEPENDS** on mass and radius of the planet/body from which it is launched, and initial launch height.

Numerical Method — Neutral Point & Minimum Projection Speed Between Two Masses

Given: Spheres M & 4M, Distance 6R → Find Neutral Point N: $G\,M\,m / r^2 = 4\,G\,M\,m / (6R-r)^2 \Rightarrow r = 2R$ → Energy Conservation: $E_{surface} = E_N$ ($v_N \approx 0$) → $\mathbf{v_{min} = \sqrt{(3\,G\,M / 5\,R)}}$

6. EARTH SATELLITES — ORBITAL SPEED, PERIOD & ENERGY

MUST KNOW

Circular Orbital Speed (v_o) & Time Period (T)

EXAM FAVOURITE

$$v_o = \sqrt{[G M_E / (R_E + h)]} \quad | \quad T = 2\pi (R_E + h)^{3/2} / \sqrt{(G M_E)}$$

Near-Earth Satellite ($h \ll R_E$): $v_o = \sqrt{(g R_E)} \approx 7.92\ km\ s^{-1}$

Near-Earth Time Period: $T_0 = 2\pi \sqrt{(R_E / g)} \approx 84.6\ min$

Escape vs Orbital Speed: $v_e = \sqrt{2} \cdot v_o \approx 1.414\ v_o$
(+41.4% speed increase needed to escape)

Geostationary Satellite: $T = 24\ hrs$, $h \approx 35{,}800\ km$ (orbits west-to-east in equatorial plane).

Energy Distribution of Orbiting Satellite (Radius r = R_E + h)

MUST KNOW

$$K = +(G M m) / (2r) \quad | \quad U = -(G M m) / r \quad | \quad E_{total} = K + U = -(G M m) / (2r)$$

Energy Relationships: $K = -E_{total}$ | $U = 2 E_{total}$ | $U = -2K$

Binding Energy (B.E.): Energy needed to remove satellite to $\infty = -E_{total} = +(G M m) / (2r)$

7. GRAVITATIONAL FIELD & POTENTIAL FOR SPHERICAL SHELLS & SOLIDS

SPECIAL CASE

Type of Body	Location	Gravitational Field (E)	Gravitational Potential (U)
Thin Uniform Spherical Shell (Mass M, Radius R)	Outside ($r \geq R$)	$E = G M / r^2$	$U = -G M / r$
	Inside ($r < R$)	$E = 0$ (Shielding effect)	$U = -G M / R$ (Constant)
Uniform Solid Sphere (Earth Model)	Outside ($r \geq R$)	$E = G M / r^2$	$U = -G M / r$
	Inside ($r < R$)	$E = (G M / R^3) \cdot r$ ($E \propto r$)	$U = -(G M / 2R^3) \cdot (3R^2 - r^2)$ Center ($r=0$): $U_{center} = 1.5\ U_{surface}$

Numerical Method — Energy Required to Transfer Satellite Between Orbits

Given: Orbit 1 ($r_1 = 2R_E$) → Orbit 2 ($r_2 = 4R_E$) → $\Delta E = E_f - E_i = [-G M m / (2 \cdot 4R_E)] - [-G M m / (2 \cdot 2R_E)]$ → $\Delta E = +G M m / (8 R_E) = (m g R_E) / 8$ → $\Delta K = -\Delta E$ | $\Delta U = +2\Delta E$

CONCEPT CLEARING — Weightlessness in Orbit:
An astronaut inside an orbiting satellite experiences weightlessness NOT because gravity is zero in space, but because both the astronaut and spaceship are in a state of continuous **free fall** toward Earth with acceleration $a = g(h)$, making normal reaction force $N = 0$.